Safety Requirements Specification (SRS)

Hybrit Pilot Plant

REV	DATE	APPROVED	DESCRIPTION OF CHANGE
	5/20/2020		

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1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the general and specific safety requirements specification (SRS) for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project in accordance with clause 10 of IEC 61511-1 / ISA 61511-1. This document provides functional requirements that are common to the SIS and all of its associated SIFs, as well as the functional and integrity requirements that are unique to each SIF. In this way repetition, conflicting, or missing requirements are reduced.

The Safety Instrumented System needs to achieve the following key functional objectives:

• Provide reliable protection for the plant personnel, the environment, and process equipment from adverse effects due to process upsets

• Provide protection against spurious / false trips to minimize lost production, equipment damage, and exposure to process hazards

• Provide effective and reliable means of routine inspection and testing of the SIS

The Safety Instrumented System includes all elements from the sensor to the final element including logic solver inputs, outputs, communications, and power supplies.

This document does not cover requirements for the following:

• Basic Process Control System (BPCS)

• Non-SIS Independent Layers of Protection

1.1 General Project Information

Project Identification:	001
Project Name:	Hybrit Pilot Plant
Project Description:	

1.2 Standards Used

This SRS is based on the following standards.

Document Number - Title	Revision	Revision Date	Type
IEC 61508 - Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems	2010	4/30/2010	
IEC 61511 - Functional safety: Safety Instrumented Systems for the Process Industry Sector	2017	8/31/2017	
ANSI/ISA 84.00.01 - Functional safety: Safety Instrumented Systems for the Process Industry Sector	2004	9/30/2004	

2 General SIS Requirements

This chapter describes the requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project. Additional or different requirements for individual SIFs will be documented in the appropriate SIF subsection of this document.

2.1 SIS Logic Solver Hardware Requirements

This section describes the Logic Solver hardware requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project.

SYSTEMATIC FAILURE PROTECTION	
Logic Solver SIL Capability (Systematic)	3
SIS LOGIC SOLVER FAILURE REQUIREMENTS	
Action on Logic Solver Fault Detection	Vote channel to trip
Actions to Achieve/Maintain Safe State when SIS is Degraded or Inoperative	
Logic Solver Mean Time to Fail Spurious	20
SIS Output State Combinations to Avoid	None
SIS INTERFACES TO ANY OTHER SYSTEMS	
Engineering Station	Engineering Station must be implemented in accordance with the project guidelines
BPCS	Control System must be implemented in accordance with the project guidelines
Operator Station (HMI)	Operator Stations must be implemented in accordance with the project guidelines
Historian/Sequence of Events	
Other	
SIS FUNCTIONAL REQUIREMENTS	
Independent Manual Shutdown	A pushbutton will be provided in the control room to manually initiate shutdown
Major Accident Survivability	2 hours
OPERATIONAL REQUIREMENTS	
SIS Start-up / Restart Requirements	None
LOGIC SOLVER ENVIRONMENTAL EXTREMES	
Ambient Temperature	0-40 °C
Humidity	0-80%
Electromagnetic Interference	0-10 V/m
Radio Interference	3 V/m
Electrostatic Discharge	6 kV
Area Classification	Logic solver placed outside atex zone
Lightning Protection	
Other	
NOTES	

User Defined Requirements	

2.2 SIS Application Software Requirements

This section describes the application software requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project.

<table>
<tr><th colspan="4">Application Program</th></tr>
<tr><td>General</td><td colspan="3">All configuration and programming requirements specified by the safety manual for the IEC 61508 SIL 3 capable PLC shall be followed.</td></tr>
<tr><td>Program Sequencing</td><td colspan="3">Inputs blocks should precede logic functions, output blocks should be last</td></tr>
<tr><td>Trip/Time Delays</td><td colspan="3">None</td></tr>
<tr><td>Process Modes of Operation</td><td colspan="3">The following operating modes are identified for each Safety Instrumented Functions: startup, normal operation, shutdown, failed and degraded</td></tr>
<tr><th colspan="4">Real Time Performance Requirements</th></tr>
<tr><td>CPU Capacity</td><td colspan="3">x.x GHz</td></tr>
<tr><td>Network Bandwidth</td><td colspan="3">x Mbs</td></tr>
<tr><td>Acceptable Performance with Faults</td><td colspan="3"></td></tr>
<tr><td>Specified Time all Trip Signals Received</td><td colspan="3">0m 0s 050ms</td></tr>
<tr><th colspan="4">Process Variable Validation</th></tr>
<tr><td rowspan="3">Process Variable (PV) Validation Criteria</td><td>Range</td><td>Low</td><td>High</td></tr>
<tr><td rowspan="2">Analog Signals</td><td>Under-range</td><td>Over-range</td></tr>
<tr><td>< 3.6 mA</td><td>> 21.5 mA</td></tr>
<tr><td>Action on PV Out of Range</td><td colspan="3">Vote for trip</td></tr>
<tr><td>Action on PV Excessive Rate of Change</td><td colspan="3"></td></tr>
<tr><td>Action on PV Frozen Value</td><td colspan="3">Vote for trip</td></tr>
<tr><td>Action on PV Open Circuit Detection</td><td colspan="3">Vote for trip</td></tr>
<tr><td>Action on PV Short Circuit Detection</td><td colspan="3">Vote for trip</td></tr>
<tr><th colspan="4">Monitoring Functionality</th></tr>
<tr><td>Application Program Self-Monitoring</td><td colspan="3">Included in SIL 3 capable PLC programming environment</td></tr>
<tr><td>Device Monitoring</td><td colspan="3">Implement line monitoring on all Energize to Trip functions</td></tr>
<tr><th colspan="4">Proof Testing Requirements</th></tr>
<tr><td>Proof Testing while Process is Operational</td><td colspan="3">No</td></tr>
<tr><td>Additional Field Device Test Requirements</td><td colspan="3">None expected, verify with device specific safety manual once devices are selected</td></tr>
<tr><th colspan="4">Application Program Failures</th></tr>
<tr><td>Dangerous Process States Resulting from Application Program Failure</td><td colspan="3">Instability to perform safety function task, addressed by using SIL capable PLC</td></tr>
<tr><th colspan="4">Communication Interfaces</th></tr>
<tr><td>Validation of Received Data</td><td colspan="3">Bounds checking/range checking</td></tr>
</table>

Validation of Transmitter Data	N/A, data send from SIL3 capable PLC
Measures to Limit Interface Use	N/A
REFERENCE DOCUMENTS	
Reference Type	**Document Number - Title**
	-
NOTES	
USER DEFINED REQUIREMENTS	

3 General SIF Requirements

This chapter describes the common functional requirements that generally apply to all SIFs in the Hybrit Pilot Plant project and that are not repeated for each specific SIF. Additional requirements for an individual SIF will be documented in the appropriate SIF subsection of this document.

MAINTENANCE OVERRIDE DEGRADATION REQUIREMENTS							
Voting Degradation	**Original**	**1oo1**	**1oo2**	**2oo2**	**1oo3**	**2oo3**	**3oo3**
	Degraded	Trip	1oo1	1oo1	1oo2	1oo2	2oo2
FIELD EQUIPMENT FAIL SAFE STATES							
Transmitter	Low output, e.g. 3.6 mA						
Switch	Open contacts, e.g. 0 mA						
Isolation Valve	Closed						
Vent Valve	Open						
Motor	Stop						
Relay	Open						
REFERENCE DOCUMENTS							
Reference Type	**Document Number - Title**						
	-						
NOTES							

4 Safety Instrumented Functions

The functional and integrity requirements unique to each SIF are detailed in separate subsequent sections. Each requirement is uniquely numbered so it can be traced through project development from design, through implementation to operation.

4.1 SIF List

The following Safety Instrumented Functions are documented in this Safety Requirements Specification.

4.1.1 Low Demand Service SIFs

SIF TAG	REQUIRED		SIF NAME
	SIL	RRF	
SIF-001	1	10	SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning - 254PI330 L1 low pressure on nitrogen close 254SV559 close 254SV561. Open bleed after GSS.
SIF-002	1	10	SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system - Low differential pressure 254XI330 (254PI330/ 347PI306) protect nitrogen and close 254SV559 and close 254SV561
SIF-003	1	10	SIL1 High level (347LZ302) in Absorber (347YM100) - High level (347LZ302 H1) in absorber (347YM100) close 347SV509 (USY feed water) and stop 346PU101 top gas compressor
SIF-004	2	100	SIL2 Low level (347LZ300) in Absorber (347YM100) - SIL2 Low level (347LZ300) in Absorber (347YM100) stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close
SIF-005	1	10	SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101 - High level (347LS305 H1) filter separator 347SE101 stop 346PU101 (top gas compressor)
SIF-006	1	10	SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper - Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close
SIF-007	2	100	SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps - High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps
SIF-008	1	10	SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608 - High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608
SIF-009	2	100	SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber - Low

			differential pressure 347PD339 Pump discharge and absorber (347YM100) Close 347SV671 Lean MEA to Absorber Close 347RV667 lean MEA to Absorber

5 SIF-001 SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning

SIF Identification			
SIF Tag	SIF-001	**Revision**	0
SIF Name	SIL1 Low Pressure (254PI330|L1) nitrogen system, CO2 cleaning		
SIF Description / Service	254PI330 L1 low pressure on nitrogen close 254SV559 close 254SV561. Open bleed after GSS.		
Functional Requirements			
Equipment	254PI330, 254SV559, 254SV561		
Process Safe State	If low pressure in nitrogen system shut-off valves will close	**Process Safety Time**	0h 0m 30s
Demand Source	Low pressure on nitrogen to CO2 stripper		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs								
Intended Tag	**Range (Min to Max)**		**Set/Trip Point**	**Trip Direction**	**Tolerance**	**UOM**	**Input Type**	**Basis of Set Point**
254PI330	0	15	3.8	Low	0	bar(g)	Process	Prevent backflow of process gas (CO2) to nitrogen grid

Intended Final Element Legs					
Intended Tag	**Trip Action**	**Tightness Required**	**Fail Position**	**Remote Actuated**	**Feedback Tag**
254SV559	Close / On	No	Closed	Yes	254SV559_GSS
254SV561	Close / On	No	Closed	Yes	254SV561_GSS

Input / Output Functional Relationship		Low pressure in nitrogen grid results in close 254SV559, open 243SV560 and close 254SV561	
Additional Mitigation	Check valve 254HV562		
Related Interlock	-		
Non-safety actions	open 254SV560		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements		None-Specific	
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	254SV559 FC, 254SV561 FC, 243SV560 FO
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.

DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	Reset should not override trip
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

OPERATIONAL REQUIREMENTS			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitable designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitable designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitable designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	**Document Number - Title**		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

6 SIF-002 SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system

SIF Identification			
SIF Tag	SIF-002	**Revision**	0
SIF Name	SIL1 Low differential pressure (254PI330 / 347PI306) Nitrogen system		
SIF Description / Service	Low differential pressure 254XI330 (254PI330/ 347PI306) protect nitrogen and close 254SV559 and close 254SV561		
Functional Requirements			
Equipment	347XI306 (254PI330, 347PI306) 254SV559, 254SV561		
Process Safe State	If differential pressure in nitrogen system is to low between 254PI330 / 347PI306 shut-off valves will close	**Process Safety Time**	0h 0m 30s
Demand Source	Low pressure on nitrogen to CO2 stripper		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs

Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
254PI330 - 347PI306	1	10	1	Low	0	bar(g)	Process	Prevent overfill of absorber 347YM100

Intended Final Element Legs

Intended Tag	Trip Action	Tightness Required	Fail Position	Remote Actuated	Feedback Tag
254SV559	Close / On	No	Closed	Yes	254SV559_GSS
254SV561	Close / On	No	Closed	Yes	254SV561_GSS

Input / Output Functional Relationship		Low pressure in nitrogen grid results in close 254SV559, open 243SV560 and close 254SV561	
Additional Mitigation	Check valve 254HV562		
Related Interlock	-		
Non-safety actions	open 254SV560		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements		None-Specific	
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	254SV559 FC, 254SV561 FC, 243SV560 FO
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.
DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	Reset should not override trip
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

Operational Requirements			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
Environmental Requirements			
Sensor	SIS must be suitable designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitable designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitable designed, built and installed to operate in the environmental conditions
Reference Documents			
Reference Type	**Document Number - Title**		
	-		
Notes			
Prevent backflow of process gas (CO2) to nitrogen grid			
User Defined Requirements			
Under utredning			

7 SIF-003 SIL1 High level (347LZ302) in Absorber (347YM100)

SIF IDENTIFICATION			
SIF Tag	SIF-003	**Revision**	0
SIF Name	SIL1 High level (347LZ302) in Absorber (347YM100)		
SIF Description / Service	High level (347LZ302 H1) in absorber (347YM100) close 347SV509 (USY feed water) and stop 346PU101 top gas compressor		
FUNCTIONAL REQUIREMENTS			
Equipment	347LZ302, 347SV509, 346PU101		
Process Safe State	If High level (347LZ302 H1) in absorber (347YM100) close 347SV509 (USY feed water) and stop 346PU101 top gas compressor	**Process Safety Time**	0h 0m 30s
Demand Source	High level (347LZ302 H1) in absorber (347YM100)		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

INTENDED SENSOR LEGS								
Intended Tag	**Range (Min to Max)**		**Set/Trip Point**	**Trip Direction**	**Tolerance**	**UOM**	**Input Type**	**Basis of Set Point**
347LZ302	0	100	97	High	0	%	Process	Prevent overfill of absorber 347YM100

INTENDED FINAL ELEMENT LEGS					
Intended Tag	**Trip Action**	**Tightness Required**	**Fail Position**	**Remote Actuated**	**Feedback Tag**
347SV509	Close / On	No	Closed	Yes	347SV509_GSS
346PU101	Open/Off	No	Closed	No	346PU101_OFF

Input / Output Functional Relationship		High level Absorber 347LZ302 H1 close 347SV509 and stop 346PU101	
Additional Mitigation	347LI508		
Related Interlock	-		
Non-safety actions	Close 347LC508 Regulator reboiler		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements		None-Specific	
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	347SV509 FC
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.

DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	Reset should not override trip
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

Operational Requirements			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
Environmental Requirements			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Reference Documents			
Reference Type	**Document Number - Title**		
	-		
Notes			
User Defined Requirements			
Under utredning			

8 SIF-004 SIL2 Low level (347LZ300) in Absorber (347YM100)

SIF Identification			
SIF Tag	SIF-004	**Revision**	0
SIF Name	SIL2 Low level (347LZ300) in Absorber (347YM100)		
SIF Description / Service	SIL2 Low level (347LZ300) in Absorber (347YM100) stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close		
Functional Requirements			
Equipment	347LZ300, 347SV672, 347SV673		
Process Safe State	If low level (347LZ300) in Absorber (347YM100) stop Rich MEA to Stripper	**Process Safety Time**	0h 0m 30s
Demand Source	Low level (347LZ300) in Absorber (347YM100)		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs

Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
347LZ300	0	100	5	Low	0	%	Process	Prevent low level in absorber => gas break through

Intended Final Element Legs

Intended Tag	Trip Action	Tightness Required	Fail Position	Remote Actuated	Feedback Tag
347SV672	Close / On	No	Closed	Yes	347SV672_GSS
347SV673	Close / On	No	Closed	Yes	347SV673_GSS

Input / Output Functional Relationship		High level Absorber 347LZ300 stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close	
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	Close 347RV608 control valve rich MEA		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements		None-Specific	
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	347SV672 FC, 347SV673 FC
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.

DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	Reset should not override trip
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

OPERATIONAL REQUIREMENTS			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	**Document Number - Title**		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

9 SIF-005 SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101

SIF IDENTIFICATION			
SIF Tag	SIF-005	**Revision**	0
SIF Name	SIL1 High level (347LS305) filter separator 347SE101 stop 346PU101		
SIF Description / Service	High level (347LS305 H1) filter separator 347SE101 stop 346PU101 (top gas compressor)		
FUNCTIONAL REQUIREMENTS			
Equipment	347LS305, 346PU101		
Process Safe State	High level in filter separator stops compressor	**Process Safety Time**	0h 0m 30s
Demand Source	High level (347LS305) filter separator 347SE101		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

INTENDED SENSOR LEGS								
Intended Tag	**Range (Min to Max)**		**Set/Trip Point**	**Trip Direction**	**Tolerance**	**UOM**	**Input Type**	**Basis of Set Point**
347LS305	0	0	0	High	0	%	Process	(VALUE TBD)

INTENDED FINAL ELEMENT LEGS					
Intended Tag	**Trip Action**	**Tightness Required**	**Fail Position**	**Remote Actuated**	**Feedback Tag**
346PU101	Open/Off	No	Closed	No	346PU101_OFF

Input / Output Functional Relationship		High level filter separator 347LS305 H1 stop 346PU101 top gas compressor	
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	-		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, Proof test offline, degration vote for trip	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response	Trip		
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	24 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

<table>
<tr><td>SIF Interfaces</td><td colspan="3">None</td></tr>
<tr><td>SIF Start-up / Restart Requirements</td><td colspan="3">None</td></tr>
<tr><td colspan="4">Environmental Requirements</td></tr>
<tr><td>Sensor</td><td>SIS must be suitably designed, built and installed to operate in the environmental conditions</td><td>Final Element</td><td>SIS must be suitably designed, built and installed to operate in the environmental conditions</td></tr>
<tr><td>Ambient Conditions</td><td>Non-specific</td><td>EMC Environment</td><td>SIS must be suitably designed, built and installed to operate in the environmental conditions</td></tr>
<tr><td colspan="4">Reference Documents</td></tr>
<tr><td>Reference Type</td><td colspan="3">Document Number - Title</td></tr>
<tr><td></td><td colspan="3">-</td></tr>
<tr><td colspan="4">Notes</td></tr>
<tr><td colspan="4">User Defined Requirements</td></tr>
<tr><td>Under utredning</td><td colspan="3"></td></tr>
</table>

10 SIF-006 SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper

SIF Identification			
SIF Tag	SIF-006	**Revision**	0
SIF Name	SIL1 Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper		
SIF Description / Service	Low pressure (347PI306 L3) treated gas stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close		
Functional Requirements			
Equipment	347PI306, 347SV672, 347SV673		
Process Safe State	If Low pressure (347PI306) treated gas stop Rich MEA to Stripper	**Process Safety Time**	0h 0m 30s
Demand Source	Low pressure (347PI306) treated gas		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs

Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
347PI306	0	8	2.5	Low	0	Bar(g)	Process	Prevent reverse flow of containment in piping between stripper 347YM102 to absorber 347YM100

Intended Final Element Legs

Intended Tag	Trip Action	Tightness Required	Fail Position	Remote Actuated	Feedback Tag
347SV672	Close / On	No	Closed	Yes	347SV672_GSS
347SV673	Close / On	No	Closed	Yes	347SV673_GSS

Input / Output Functional Relationship		High level Absorber 347LZ300 stop Rich MEA to Stripper -> 347SV672 close and 347SV673 close	
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	Close 347RV608 control valve rich MEA		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	347SV672 FC, 347SV673 FC
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.

DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	None
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

Operational Requirements			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
Environmental Requirements			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Reference Documents			
Reference Type	**Document Number - Title**		
	-		
Notes			
User Defined Requirements			
Under utredning			

11 SIF-007 SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps

SIF Identification			
SIF Tag	SIF-007	**Revision**	0
SIF Name	SIL2 High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps		
SIF Description / Service	High Temperature (347TI338) after lean MEA cooler close 347SV674 suction side MEA pumps		
Functional Requirements			
Equipment	347TI338, 347SV674		
Process Safe State	If High Temperature after lean MEA cooler close valve suction side MEA pumps	**Process Safety Time**	0h 0m 30s
Demand Source	High Temperature after lean MEA		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs								
Intended Tag	**Range (Min to Max)**		**Set/Trip Point**	**Trip Direction**	**Tolerance**	**UOM**	**Input Type**	**Basis of Set Point**
347TI338	0	100	70	High	0	T(C°)	Process	Prevent high temperature

Intended Final Element Legs					
Intended Tag	**Trip Action**	**Tightness Required**	**Fail Position**	**Remote Actuated**	**Feedback Tag**
347SV674	Close / On	No	Closed	Yes	347SV674 _GSS

Input / Output Functional Relationship		High Temperature (347TI338) after lean MEA cooler close 347SV674 valve suction side MEA pumps	
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	Trip (347PU104) Lean MEA pump A Trip (347PU105) Lean MEA pump B Close (347RV667) Lean MEA to Absorber Close (347SV671) Lean MEA to Absorber Close (347FC667) Lean MEA to Absorber		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response	347SV674 FC		
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard	None		
SIS Output State Combinations to Avoid	None		
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		

<table>
<tr><th colspan="4">OVERRIDE REQUIREMENTS</th></tr>
<tr><td>Startup Overrides</td><td colspan="3">Reset should not override trip</td></tr>
<tr><td>Maintenance Overrides</td><td colspan="3">No overrides acceptable</td></tr>
<tr><td>Bypass Requirements</td><td colspan="3">N/A</td></tr>
<tr><th colspan="4">OPERATIONAL REQUIREMENTS</th></tr>
<tr><td colspan="2">SIF Interfaces</td><td colspan="2">None</td></tr>
<tr><td colspan="2">SIF Start-up / Restart Requirements</td><td colspan="2">None</td></tr>
<tr><th colspan="4">ENVIRONMENTAL REQUIREMENTS</th></tr>
<tr><td>Sensor</td><td>SIS must be suitably designed, built and installed to operate in the environmental conditions</td><td>Final Element</td><td>SIS must be suitably designed, built and installed to operate in the environmental conditions</td></tr>
<tr><td>Ambient Conditions</td><td>Non-specific</td><td>EMC Environment</td><td>SIS must be suitably designed, built and installed to operate in the environmental conditions</td></tr>
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<tr><td></td><td colspan="3">-</td></tr>
<tr><th colspan="4">NOTES</th></tr>
<tr><th colspan="4">USER DEFINED REQUIREMENTS</th></tr>
<tr><td>Under utredning</td><td colspan="3"></td></tr>
</table>

12 SIF-008 SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608

SIF Identification			
SIF Tag	SIF-008	**Revision**	0
SIF Name	SIL1 High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608		
SIF Description / Service	High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608		
Functional Requirements			
Equipment	347LZ302, 347RV608		
Process Safe State	If High level (347LZ329) in MEA reboiler (347HE201) close 347RV608	**Process Safety Time**	0h 0m 30s
Demand Source	High level (347LZ329) in MEA reboiler (347HE201)		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs

Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
347LZ329	0	100	95	High	0	%	Process	Prevent overfill

Intended Final Element Legs

Intended Tag	Trip Action	Tightness Required	Fail Position	Remote Actuated	Feedback Tag
347RV608	Close / On	No	Closed	Yes	347RV608_GSS

Input / Output Functional Relationship		High level (347LZ329 H1) in MEA reboiler (347HE201) close 347RV608	
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	-		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Matrix		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	347SV509 FC
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.

DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	Reset should not override trip
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

OPERATIONAL REQUIREMENTS			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	**Document Number - Title**		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

13 SIF-009 SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber

SIF Identification			
SIF Tag	SIF-009	**Revision**	0
SIF Name	SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber		
SIF Description / Service	Low differential pressure 347PD339 Pump discharge and absorber (347YM100) Close 347SV671 Lean MEA to Absorber Close 347RV667 lean MEA to Absorber		
Functional Requirements			
Equipment	347PD339, 347SV671, 347RV667		
Process Safe State	If differential pressure in nitrogen system is to low between 254PI330 / 347PI306 shut-off valves will close and bleed will open	**Process Safety Time**	0h 0m 30s
Demand Source	Low pressure on nitrogen to CO2 stripper		
Demand Rate	0.000E+000 per year	**Demand Mode**	Low
Operating Modes	Normal, Start-up, Shutdown, Failed, Degraded		
Manual Shutdown	Yes		
Protection Method	DeEnergizeToTrip		
Trip Reset	Manual	**Local/Remote**	Local
Trip Reset Description			
Response Time SIF	10 seconds		
Maximum Part Response Time	**Sensor**	**Logic Solver**	**Final Element**
	msec	msec	msec
Mission Time	15 years		

Intended Sensor Legs

Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
347PD339	0	250	50	Low	0	mbar(g)	Process	

Intended Final Element Legs

Intended Tag	Trip Action	Tightness Required	Fail Position	Remote Actuated	Feedback Tag
347SV671	Close / On	No	Closed	Yes	347SV671_GSS
347RV667	Close / On	No	Closed	Yes	347RV667_GSS

Input / Output Functional Relationship		SIL2 Low differential pressure 347PD339 Pump discharge and absorber (347YM100) close 347SV671 lean MEA to Absorber	
Additional Mitigation	347FI667 & trips pump on no flow		
Related Interlock	-		
Non-safety actions	-		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements		-	
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degration vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	**Integrity Level**	**Required RRF**
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		

SIF FAILURE REQUIREMENTS	
Mean Time to Fail Spurious	1 per 10 year(s)
Sensor Failure Response	Vote to trip
Logic Solver Failure Response	Vote channel to trip
Final Element Failure Response	
Maximum Restore to Service Time	24 hours
Startup/Restart Time	8 hours
Concurrent Safe Process States Hazard	None
SIS Output State Combinations to Avoid	None
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.

DESIRED PROOF TEST INTERVALS	
Sensor Part	12 months
Logic Solver Part	12 months
Final Element Part	12 months
OVERRIDE REQUIREMENTS	
Startup Overrides	Reset should not override trip
Maintenance Overrides	No overrides acceptable
Bypass Requirements	N/A

OPERATIONAL REQUIREMENTS			
SIF Interfaces	None		
SIF Start-up / Restart Requirements	None		
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	**Final Element**	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	**EMC Environment**	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	**Document Number - Title**		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

14 Abbreviations and Definitions

14.1 Abbreviations

BMS	Burner Management System
BPCS	Basic Process Control System
E/E/PE	Electrical/Electronic/Programmable Electronic
EMC	Electro-Magnetic Compatibility
FAT	Factory Acceptance Testing
FPL	Fixed Program Language
FVL	Full Variability Language
HFT	Hardware Fault Tolerance
HMI	Human Machine Interface
IEC	International Electrotechnical Commission
IPL	Independent Protection Layers
ISA	Instrumentation, Systems and Automation Society
LVL	Limited Variability Language
MooN	“M” out of “N”
MTTFS	Mean Time To Fail Spurious
MRT	Mean Repair Time
PFD	Probability of Failure on Demand
PFDavg	Average Probability of Failure on Demand
PFH	Probability of a Dangerous Failure per Hour
PLC	Programmable Logic Controller
PTC	Proof Test Coverage
PTI	Proof Test Interval
RRF	Risk Reduction Factor
SAT	Site Acceptance Test
SERH	Safety Equipment Reliability Handbook
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SLC	Safety Life Cycle
SRS	Safety Requirements Specification

14.2 Definitions

Low Demand Mode	the demand is infrequent and the demand frequency is less than half the proof test frequency, e.g. proof tests are accounted for
High Demand Mode	the demand is somewhat frequent, the demand frequency is greater than half the proof test frequency, and the worst case diagnostic time interval is an order of magnitude smaller than the demand interval, e.g. proof tests are not accounted for, diagnostics are accounted for
Continuous Demand Mode	the demand is frequent, the demand frequency is greater than half the proof test frequency, and the worst case diagnostic time interval is not an order of magnitude smaller than the demand interval, e.g. proof tests are not accounted for, diagnostics are not accounted for

15 Disclaimer, Assumptions, Equipment Data

15.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

15.2 Assumptions Safety Requirements Specification

This safety requirements specification document is generated based on the selections the user made during the SIL selection and SIL verification activities in combination with specific safety requirements specification entries on both project and SIF level basis.

The cause and effect diagram that is created as part of the SIF Functional Relationship only depicts the actions to be taken for the specific SIF under consideration. If multiple SIFs initiate based on a specific sensor group and/or operate the same final element group this will not be reflected in these individual cause and effect diagrams. A complete cause and effect diagram taking into consideration all Safety Instrumented Functions will show these commonalities assuming that the user has correctly identified identical groups and has used the reuse feature in the SILver tool to identify these identical groups. A complete, i.e. project level, cause and effect diagram can be generated independent of this report.

The position of this safety requirements specification document within the overall safety lifecycle deviates from the lifecycles published in the functional safety standards. Typically the SRS phase is located between SIL selection and Conceptual Design, i.e. SIL verification. The required information in the SRS however covers information developed in the SIL selection phase as well as in the Conceptual Design / SIL verification phase. For example specific application level diagnostic requirements like external comparison of analog signals or the implementation of partial valve stroke testing are determined during the SIL verification but also need to be documented in the safety requirements specification document. Therefore the exSILentia® software allows users to create Process SRS as well as a Design SRS document to better conform to normal work practices.